

# Attack-Aware Market Making: Adversarial Co-Training with Economically-Constrained Spoofing Agents

Literature Review

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## Introduction

Electronic financial markets rely on automated market makers: algorithmic agents that continuously quote bid and ask prices for liquidity provision and price discovery [1], [2]. Deep reinforcement learning (DRL) is increasingly applied to this role, with agents trained to balance spread capture against inventory risk by conditioning on the instantaneous limit-order-book (LOB) state [3], [4]. Section 1 establishes the microstructural foundations on which these formulations rest: inventory-risk and adverse-selection models, order-flow imbalance dynamics, and regime-dependent volatility and liquidity provision. Section 2 surveys DRL market-making algorithms, multi-agent training dynamics, and simulation platform trade-offs, with particular attention to the market-replay constraint of JaxMARL-HFT [5], [6].

The same LOB conditioning that makes DRL market makers effective exposes them to adversarial manipulation. Spoofing, placing large orders with no intent to execute, is illegal under the Dodd-Frank Act and was implicated in the 2010 Flash Crash [7]; a spoofed order shifts apparent depth, inducing an unaware agent to revise quotes in the adversary’s intended direction [8]. Section 3 examines spoofing’s economic structure, the empirical detection literature, and its dual-use implications. Section 4 covers adversarial co-training precedents, robust MDPs, and the auxiliary-task learning framework underlying the proposed spoof-detection head. Section 5 synthesises all four literatures to identify what is established, what is contested, and what the proposed system addresses.

## 1 Market microstructure foundations

### 1.1 The limit order book

The limit order book is the data structure through which more than half of the world’s financial markets match buyers and sellers [2]. At any instant, the LOB records all outstanding limit orders

on both sides, organised by price level and, within a level, by time of arrival. Three fundamental order types are submitted: limit orders post visible interest at a specified price or better; market orders execute immediately against the best available counterparty; cancellations withdraw a previously submitted limit order. The best bid and best ask are the highest outstanding buy price and lowest outstanding sell price, the spread is their difference and the mid-price their midpoint.

Most contemporary venues match orders under strict price–time priority: incoming aggressive orders match against resting orders at the best price first, and within a price level orders fill in arrival order. Cancellations alter visible depth but not the priority of remaining orders. That limit orders must wait their turn but can be withdrawn instantly (sometimes sub-milliseconds) is what makes spoofing operationally feasible: a large limit order contributes to the visible depth signal up to its cancellation [2].

## 1.2 Market making: inventory risk and adverse selection

The market-making literature is organised around two complementary sources of risk: inventory risk, the possibility that the dealer’s position moves adversely before being offloaded, and adverse selection, the risk of trading against better-informed counterparties. Both are directly relevant to spoofing, which exploits a market maker’s rational response to perceived information in order flow. Foucault, Pagano and Röell (2013) provide the standard textbook synthesis [9].

**Inventory-risk models.** Ho and Stoll (1981) provide the foundational treatment, deriving optimal bid and ask prices through dynamic programming for a single dealer facing stochastic order arrivals; reservation prices skew away from the midpoint when inventory is non-zero, in the direction that discourages further accumulation [10]. Avellaneda and Stoikov (2008) extend the framework to the high-frequency setting, deriving a closed-form continuous-time quoting strategy in which the agent quotes symmetrically around an inventory-adjusted reservation price [3]. The core mechanism is intuitive: rather than quoting around the mid, the agent quotes around a *reservation price* shifted from the mid in proportion to current inventory, with the spread widening as residual time horizon and volatility grow. Formally, with mid-price  $S_t$ , inventory  $q_t$ , risk-aversion coefficient  $\gamma$ , return volatility  $\sigma$ , terminal time  $T$ , and order-arrival sensitivity  $k$ , the A–S reservation price and total quoted spread are

$$r(S_t, q_t, t) = S_t - q_t \gamma \sigma^2 (T - t), \quad \delta^a + \delta^b = \gamma \sigma^2 (T - t) + \frac{2}{\gamma} \ln\left(1 + \frac{\gamma}{k}\right), \quad (1)$$

with bid and ask posted symmetrically around  $r$ . The skew of  $r$  in proportion to  $q_t$  is the inventory-management mechanism DRL agents are typically calibrated against, and A–S is the analytical benchmark in subsequent work [5]. The A–S exponential order-arrival intensity is convenient but has not been systematically tested against tick-constrained markets. Both Ho–Stoll and A–S model the market maker in isolation against an exogenous mid-price diffusion — an assumption violated wherever the market maker’s quotes are observed and reacted to, including, critically, by an adversarial spoofer.

**Adverse-selection models.** Glosten and Milgrom (1985) derive equilibrium quotes for a competitive market maker facing order flow composed of uninformed and informed traders, with the spread widening as the perceived proportion of informed traders rises [11]. Kyle (1985) formalises a single strategic insider trading against a market maker who sets prices as a linear function of aggregate order flow, with the resulting price-impact coefficient  $\lambda$  reflecting the informativeness of net order flow [12]. The two papers differ in whether informed trading is dispersed or concentrated, but both make the market maker’s response a function of the inferred informativeness of order flow.

This adverse-selection channel is the mechanism through which spoofing operates: by injecting flow that mimics the signature of informed trading, the spoofer induces the market maker to update quotes as if facing genuine adverse selection. Both Glosten–Milgrom and Kyle assume the market maker cannot distinguish informed from uninformed order flow; the spread is the only response.

The empirical evolution of this line has produced operational metrics estimating adverse-selection intensity from observed order flow. Easley, Kiefer, O’Hara and Paperman (1996) introduced the Probability of Informed Trading (PIN), recovered via maximum-likelihood estimation under a structural mixture-of-Poissons model [13]. Easley, López de Prado and O’Hara (2012) extended this to high-frequency settings with the Volume-synchronised PIN (VPIN), aggregating trades into equal-volume buckets and using the absolute volume imbalance as a real-time proxy for flow toxicity [14]. The framework has been actively contested. Andersen and Bondarenko (2014) argue that VPIN peaked *after* rather than before the Flash Crash, that its short-run volatility predictiveness is primarily a mechanical consequence of trading intensity, and that the bulk-volume trade-classification scheme underperforms transaction-level classification [15]; Easley, López de Prado and O’Hara (2014) defend the broader framework in a rejoinder, and the exchange has not been definitively resolved [16]. For the present setting, PIN/VPIN are functional analogues of the discriminative representation a learned spoof-detection head would compute — but they aggregate over directional volume imbalance without distinguishing spoofing’s cancellation-heavy signature, so whether a representation specifically trained to detect spoofing can outperform a PIN/VPIN-style aggregate signal remains open.

### 1.3 Price impact, order flow, and cancellations

The relationship between order-flow imbalance and price movement is central to the spoofing problem. Cont, Kukanov and Stoikov (2014), hereafter CKS, show using NYSE TAQ data on fifty U.S. stocks that the order-flow imbalance (OFI) — the net signed change in best-bid and best-ask depth over a short interval — predicts short-horizon mid-price change substantially better than trade volume, with an approximately linear relationship and a slope inversely proportional to depth at the best quote [17]. OFI signs each depth change at the touch by its expected price-pressure direction (bid-increase or ask-decrease pushes the mid up, and conversely). Formally, for the  $n$ -th order-book event with best prices  $P_n^B, P_n^A$  and posted depths  $q_n^B, q_n^A$ , the per-event OFI contribution is

$$e_n = q_n^B \mathbb{1}_{\{P_n^B \geq P_{n-1}^B\}} - q_{n-1}^B \mathbb{1}_{\{P_n^B \leq P_{n-1}^B\}} - q_n^A \mathbb{1}_{\{P_n^A \leq P_{n-1}^A\}} + q_{n-1}^A \mathbb{1}_{\{P_n^A \geq P_{n-1}^A\}}, \quad (2)$$

with  $\text{OFI}_{[t_{k-1}, t_k]} = \sum_n e_n$  and CKS specification  $\Delta P_k = (c/\bar{D}_k) \cdot \text{OFI}_k + \epsilon_k$ , where  $\bar{D}_k$  is average best-quote depth. OFI drives the primary forecast signal in DRL market makers, and the inverse-depth coefficient implies that a spoofed order’s price impact varies systematically with market conditions.

The CKS specification warrants scrutiny. The linearity is empirical, estimated on a specific NYSE sample; whether it holds across instruments, venues, and regimes is not established. CKS include cancellations among the three event types studied, but the cancellation population aggregates manipulative and bona-fide cancellations, with the latter dominating by base rate. Khomyn and Putniņš (2021) document that 97% of limit orders in U.S. equity markets are cancelled before execution, the bulk of which they attribute to legitimate liquidity provision rather than manipulation [18]; Comerton-Forde and Putniņš (2014) provide a complementary upper bound from the opposite direction, estimating that approximately 1% of closing prices in their sample show evidence of manipulation [19]. Using the aggregate coefficient as a per-spoof impact estimate therefore conflates a population-average effect with the effect on the manipulative subpopulation specifically, with the sign and magnitude of any resulting error uncharacterised. The difficulty of

separating the two populations is inherent rather than incidental: if manipulative cancellations were straightforwardly distinguishable from ordinary ones by their observable signatures, spoofing would be trivially detectable.

Hasbrouck and Saar (2009), studying one hundred NASDAQ stocks on INET, document that over one-third of nonmarketable limit orders are cancelled within two seconds — a population they term “fleeting orders” — proposing price-chasing and latent-liquidity search as dynamic-trading-strategy explanations, but not directly estimating their price impact [20]. CKS and Hasbrouck–Saar together establish that rapid cancellation is prevalent and that cancellation events carry measurable price impact, while leaving open whether spoofed cancellations operate through the same mechanism as the broader population.

The depth–volatility relationship adds further complication. Brogaard et al. (2018) document that HFTs typically supply liquidity during single-stock extreme price movements but switch to net liquidity demand under simultaneous multi-stock extremes, interpreting the latter as a limit of liquidity provision under correlated stress rather than HFT-caused movement [21]. This interacts with the CKS inverse-depth coefficient: a spoofed order of fixed size produces materially different apparent imbalance, and price impact, in calm versus stressed regimes. Whether this translates into different attack efficacy against a DRL market maker is empirically open.

#### 1.4 Volatility regimes and conditioning

The microstructure dynamics surveyed above are not stationary in time. Engle (1982) and Bollerslev (1986) established the autoregressive conditional heteroskedasticity framework, demonstrating that return volatility clusters in distinct high- and low-variance periods [22], [23]. Hamilton (1989) formalised the clustering as a Markov-switching process in which the unobserved regime indicator follows a first-order Markov chain [24]. Corsi (2009) introduced the Heterogeneous Autoregressive model of Realized Volatility (HAR-RV), now the standard forecasting model for high-frequency volatility [25].

The regime-conditioning case for financial decision-making is anchored by Ang and Bekaert (2002), who solve a dynamic portfolio choice problem under a regime-switching investment opportunity set and document that ignoring regime dependence imposes economically significant costs; their finding that correlations and volatilities co-move in bear regimes has direct microstructural analogues [26]. Brogaard et al. (2018) document the most directly relevant: HFT liquidity provision is regime-dependent [21], motivating the hypothesis that depth, and hence the CKS inverse-depth scaling, may itself be regime-dependent. The VIX — the CBOE’s implied-volatility index — is the most widely used contemporaneous market-stress indicator and a plausible candidate for an explicit regime-conditioning input to a market-making policy [27].

The reinforcement-learning side is provided by the contextual MDP (CMDP) framework of Hallak, Di Castro and Mannor (2015) [28]. A CMDP augments the standard MDP tuple with a context space  $\mathcal{C}$  on which dynamics and rewards depend; the policy becomes  $\pi(a \mid s, c)$  rather than  $\pi(a \mid s)$ , and the learning problem is to recover a policy that performs well across the context distribution. In the present application  $c$  is a regime indicator. The Spooner–Savani (2020) formulation by contrast holds the volatility coefficient fixed across training and the market maker has no regime-state input.

Two strands of work add operational substance to the regime-conditioning case. Adams and MacKay [29] introduce Bayesian online change-point detection (BOCPD), recovering the run-length distribution since the last regime shift via a forward message-passing algorithm without a pre-fitted model. Tsaknaki, Lillo and Mazzarisi [30] apply a score-driven BOCPD extension to NASDAQ order-flow data, reporting that regime-aware online prediction of order flow and market impact significantly outperforms i.i.d. baselines—direct evidence that regime shifts at high

frequency are both detectable in real time and operationally informative. On the microstructure side, Brogaard et al.’s [21] HFT-liquidity switching under correlated stress is the canonical regime-conditional liquidity-provision result; the channel through which depth, and hence the CKS inverse-depth scaling, varies across regimes has not been formalised in market-making policy design.

Two observations close this discussion. Although regime-switching econometrics and contextual RL are individually mature, their combination in a market-making policy conditioned on a volatility-regime indicator has not been examined in the literature surveyed here. The regime indicator itself is a design choice with no high-frequency-market-making consensus: realised volatility on a trailing window is observable in real time; the VIX is forward-looking; Hamilton-style filtered probabilities require a fitted model; BOCPD run-length posteriors require a conjugate observation model.

## 2 Deep reinforcement learning for market making

### 2.1 Formulation and algorithms

Reinforcement learning frames market making as a sequential decision process: state is the featurised LOB, action a pair of quote offsets relative to mid, reward a per-step PnL adjusted for inventory risk. The standard reward combines realised profit, a variance penalty, and an inventory penalty; the variance penalty discourages slow inventory unwinding and the inventory penalty disincentivises large open positions [8]. The reward choice is itself consequential, but the post-A–S literature has not systematically compared reward specifications under adversarial conditions.

Proximal Policy Optimisation, introduced by Schulman et al. (2017), is the dominant policy-gradient algorithm in contemporary DRL market making [31]. PPO addresses a failure mode of vanilla policy-gradient updates: a single large update moves the policy too far from the data-generating distribution, collapsing subsequent learning. The fix is a clipped surrogate objective bounding the per-update policy ratio. Formally, with policy ratio  $r_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\theta_{\text{old}}}(a_t | s_t)$ , advantage  $\hat{A}_t$ , and clip parameter  $\epsilon$  (typically 0.1 or 0.2):

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[ \min \left( r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]. \quad (3)$$

PPO is on-policy, so experience from rare events such as spoofing attacks is discarded after each update; off-policy alternatives such as Soft Actor-Critic could exploit rare episodes through replay, but introduce importance-weighting complications under a non-stationary opponent. The PPO default in financial DRL is a convention inherited from general RL rather than a domain-specific finding.

### 2.2 DRL market makers in practice

Ganesh et al. (2019) demonstrate in a multi-agent dealer simulation that a DRL market maker learns to skew quotes asymmetrically in response to inventory, recovering A–S-style behaviour as a limiting case [4]. The recovery is often cited as evidence that DRL agents learn economically meaningful behaviour, but warrants scrutiny: the reward function includes an explicit inventory penalty that mechanically produces A–S-like skewing, and whether the agent has learned anything beyond what the reward directly incentivises is not addressed. The setting is a synthetic dealer market without manipulation, against a fixed population of liquidity-taking counterparties rather than an adaptive adversary — a stationary environment with no strategic opponent.

The broader DRL market-making literature extends this baseline along several dimensions, but none toward adversarial robustness. Guéant and Manziuk (2019) scale DRL market making

to multi-asset corporate-bond universes via a model-based actor-critic algorithm where the A–S HJB equations become intractable [32]. Gašperov and Kostanjčar (2021) enrich the market-maker’s state with predictive order-flow signals [33]; their (2022) work pairs DRL with a multivariate Hawkes-process LOB simulator capturing order-arrival self-excitation [34]. None includes an adversarial counterparty or a detection channel; the structural vulnerability — state representations contain no signal distinguishing spoofed depth from bona-fide depth — is common to the line. This is the vulnerability that adversarial training (Section 4) attempts to address.

**State representations and learned features.** Orthogonal to the policy-side advances, deep-learning approaches to representing LOB state have developed in parallel. Sirignano [35] introduces a spatial neural-network architecture exploiting local structure across LOB price levels, with predictive gains over fully-connected baselines at depth. Sirignano and Cont [36] report that a sufficiently large network trained on order-flow histories recovers a universal, asset-stationary price-formation relation across U.S. equities. Briola, Turiel and Aste [37] compare CNN-LSTM, attention-augmented LSTM and MLP architectures on the same LOB task and find simpler MLPs match or exceed elaborate ones, suggesting effective spatio-temporal dimensionality lower than the data structure implies. Kolm, Turiel and Westray [38] show off-the-shelf networks on stationary order-flow-imbalance features outperform networks trained on raw LOB states; Kolm and Ritter [39] survey RL applications in finance more broadly. None of this line is integrated into a DRL market-making policy in the works surveyed above, leaving the question of what shared-encoder architecture best supports both a quoting policy and an auxiliary detection head an open design problem.

### 2.3 Multi-agent training dynamics

In multi-agent settings, Independent PPO (IPPO) treats each agent’s opponents as part of the environment, training each policy independently against the rollouts they generate. Schroeder de Witt et al. (2020) demonstrate that despite violating the stationarity assumption underlying single-agent policy-gradient theory, IPPO matches or exceeds joint-learning approaches on the cooperative StarCraft Multi-Agent Challenge benchmark [40]. This cooperative result does not transfer cleanly to competitive zero-sum settings: when both agents update simultaneously and independently, the environment distribution shifts as the other improves, and joint training can cycle indefinitely between dominated policies without converging.

Documented mitigations include alternating policy freezes, fictitious self-play (training against an averaged history of past opponents), and league-based training (a population with diverse pairings). League-based training is the most theoretically grounded for zero-sum settings as it approximates a population-level Nash equilibrium but requires substantially more compute and has not been applied to financial RL. Whether alternating freezes produce stable policies in market-maker-versus-spoofers games is open.

### 2.4 Simulation platforms and their limitations

On-market reinforcement learning is economically infeasible for the agents considered here: PPO requires millions of environment interactions to converge, each incurring transaction costs and market impact in a live setting [5], [6]. All prior DRL market-making work therefore relies on simulation, and simulator quality is a first-order determinant of supportable conclusions.

Frey et al. (2023) introduce JAX-LOB, a GPU-accelerated LOB simulator with substantial per-message speedups over CPU implementations [6]. Mohl et al. (2025) extend this to multi-agent training as JaxMARL-HFT, reporting up to a  $240\times$  end-to-end training speedup and demonstrating two-player IPPO training on one year of LOB data covering 400 million orders, pairing an order-execution agent with a market maker [5]. The demonstration validates capacity

for heterogeneous two-agent training, but not specifically the adversarial market-maker-versus-attacker configuration the spoofing problem requires.

JaxMARL-HFT is one of several open platforms. ABIDES (Byrd, Hybinette and Balch, 2020) is an agent-based interactive discrete-event simulator modelled on the NASDAQ ITCH/OUCH protocols [41], exposed to the standard RL stack via ABIDES-Gym (Amrouni et al., 2021) [42]. PyMarketSim (Mascioli et al., 2024) is an agent-based environment built around private valuations and asymmetric information [43]. The platforms differ principally in price dynamics: ABIDES and PyMarketSim support Interactive Agent-Based Simulation (IABS), in which market state is generated by simulated agents and is endogenously reactive. The choice between JaxMARL-HFT and the IABS platforms is a throughput-versus-causal-fidelity trade-off: GPU acceleration enables hyperparameter sweeps at scales agent-based platforms cannot match, while IABS supports endogenous price reaction that replay cannot. JaxMARL-HFT is chosen here; the replay constraint is addressed in the formulation of the adversary itself.

Both JAX-LOB and JaxMARL-HFT rest on a market-replay assumption: the historical order book evolves along its recorded path regardless of agent actions. In pure replay, the adversary’s injected orders modify the apparent LOB state but do not causally propagate — injected depth does not move prices endogenously. An adversary whose orders cannot move the mid-price is functionally a state-perturbation agent regardless of economic framing. Spoofing’s operational logic is to induce real price movement through false signals; adversarial training under pure replay therefore tests robustness to *state perturbation*, not to *spoofing* as the microstructure literature defines it.

The fully causal alternative requires modelling the market’s response to spoofed volume, on which no consensus model exists. How an adversary should be formulated within replay’s constraints (so the training signal remains informative despite the absence of endogenous price impact) is an open design problem.

### 3 Spoofing and market manipulation

#### 3.1 Definition, legal framework, and enforcement

Spoofing is the placement of large, visible orders with no intention of execution, typically cancelled before they fill, designed to create a misleading impression of supply or demand. Layering is a variant in which multiple spoofed orders are stacked across adjacent price levels to amplify the apparent depth imbalance.

In the United States, commodities-market spoofing was criminalised under Section 747 of the Dodd–Frank Act (2010), and equity-market spoofing is prosecuted by the SEC and FINRA under the Securities Exchange Act of 1934. The EU prohibits it under the Market Abuse Regulation (596/2014) and MiFID II.

The enforcement record provides the only publicly available data on the economics of spoofing. The largest recent equity-spoofing action is *SEC v. Lek Securities and Avalon FA Ltd.* (S.D.N.Y., jury verdict November 2019), in which Avalon generated over \$25 million in illicit profits from layering U.S. equities between December 2010 and September 2016 [44]. The record suffers from survivorship bias and unrepresentative institutional context; it serves as a plausible-range anchor, not a population estimate.

The 2010 Flash Crash remains the canonical case study for the destabilising effect of manipulative order flow on an automated market. Kirilenko et al. (2017) use transaction-level CFTC data to characterise the “hot-potato” dynamic among HFTs during the crash, in which rapid net-position rotation generated apparent volume without absorbing the underlying imbalance,

an effect subsequent work has interpreted as exacerbating rather than initiating the crash’s severity [7]. Together with the enforcement record, this supports feasibility: spoofing can profit despite the realised costs of detection and prosecution.

### 3.2 Economic models of spoofing

Cartea, Jaimungal and Wang (2020) formulate spoofing as a stochastic optimal-control problem, in which an investor liquidating a position trades off manipulation gains against the regulator’s expected fine, with the optimal control characterised by HJB equations [45]. Their structural decomposition separates spoofer costs into an accidental-fill cost (spoof orders are occasionally filled and must be unwound at adverse prices) and a regulatory cost (a smooth penalty proportional to spoofed volume, parameterised by an intensity coefficient). This is the canonical structure for per-action cost models of spoofing in the equity-microstructure literature. Cartea et al. also decompose spoofer PnL into mid-price trending toward the spoofed direction (intended) and round-trip profits on inadvertently filled orders (unintended).

Cartea, Chang and García-Arenas (2023) extend the line to learning agents, calibrating to NASDAQ order-book data and characterising the conditions under which an RL market-making algorithm will discover order-book manipulation as an emergent strategy, with spoofing the special case where the agent prefers its manipulative orders not fill [46]. Two findings extend the 2020 framework: manipulation rises with the agent’s inventory-risk tolerance, suggesting it partly substitutes for A–S-style inventory management; and two learning market makers can learn to coordinate manipulation. The 2020-to-2023 line traces progressive endogenisation, from hand-coded controller to emergent strategy. A common limitation: the manipulating agent is a market maker, not an external adversary; the setting in which an external adversary learns to spoof a separate market maker is unexamined.

**Transporting the cost decomposition to replay-based simulation.** Cartea et al.’s accidental-fill plus regulatory-penalty decomposition is defined for an adversary acting against a causally responsive market: accidental fills require real counterparties; the regulatory penalty is calibrated to actual exchange surveillance. In a replay-based simulator (Section 2), neither cost arises in its native form. The decomposition nonetheless transports as a per-action shadow-cost structure: the accidental-fill cost can be retained directly, since replay does match injected orders against historical liquidity at the touch; the regulatory penalty becomes a hand-set intensity coefficient chosen to constrain spoof volume below an economically plausible profile. The calibration is therefore normative—the cost structure encodes what a real adversary would face, not what this in-simulator adversary faces, and any empirical claim about adversary effectiveness inherits this distinction.

### 3.3 Detection approaches and their limitations

The empirical characterisation of spoofing signatures is provided by Do and Putniņš (2023), who use a comprehensive global sample of prosecuted spoofing cases with out-of-sample cross-validation to identify the most diagnostic features: unbalanced order-book quotes, abnormally high order activity, abnormal cancellation rates, and cyclical patterns in market depth and cancellations [47]. A machine-learning classifier on these features achieves high out-of-sample accuracy.

The results require careful interpretation. The classifier is trained and evaluated on *prosecuted* cases; those flagrant enough to be detected and prosecuted. The base rate of spoofing in all order-book events is extremely low, and high accuracy on a balanced prosecuted dataset does not translate to acceptable false-positive rates on the full order stream. A 0.1% legitimate-event



misclassification rate would produce far more false positives than true positives across a full trading day, rendering the detector operationally useless without post-filtering.

Wang and Wellman (2020) frame detection as an adversarial learning problem with three components: a generator that adapts manipulation order streams to resemble normal trading; a discriminator distinguishing manipulated from bona-fide flow; and an agent-based simulator evaluating whether the adapted output still produces intended price impact [48]. Where Do and Putniņš ask which features distinguish observed manipulation, Wang and Wellman ask whether a manipulator can adapt to be indistinguishable. The framework produces detection rules robust to adversarial evasion, but the detection signal is external to any specific market maker: the representation is not shared with a downstream policy.

A structural observation spans both detection lines: a tension between spoofing being *effective* (large enough to shift the depth signal and move prices) and *invisible* (indistinguishable from normal flow). Spoofing must produce large visible orders, those orders must be cancelled before execution (producing abnormal cancellation rates by construction), and the spoofer must profit through a cyclical inject-cancel-trade pattern. The structural constraints on profitable spoofing may guarantee a minimum level of detectability, but the location of this effectiveness–detectability frontier is unknown.

### 3.4 Dual-use considerations

A literature review of methods to make market makers robust to spoofing passes through the literature on how to construct spoofing agents, including the learned spoofing agents of Section 3.2. The dual-use character is acknowledged in the broader AI-ethics literature on malicious applications of machine learning [49], which identifies offensive–defensive co-evolution as a recurring pattern in security-adjacent ML. Whether a published adversarial spoofing agent — even one constrained to observation-space perturbation in a replay-based simulator, with per-action costs calibrated to make spoofing operationally unattractive — contributes meaningfully to operational manipulation risk is a matter for institutional oversight rather than the literature. What the literature does show is that adversarial-MM publications have not engaged with the dual-use literature explicitly, an absence conspicuous given the operational character of the systems modelled.

## 4 Adversarial and robust approaches

### 4.1 Adversarial reinforcement learning: general methods

The canonical reference for adversarial co-training as a robustness mechanism in deep RL is Pinto et al. (2017), who introduce Robust Adversarial Reinforcement Learning (RARL) as a two-player zero-sum game between a protagonist policy and an adversary applying disturbance forces to the system [50]. The formulation is inspired by  $H_\infty$  control: modelling errors and training-deployment discrepancies are both treated as adversarial disturbances. RARL reports robustness gains across MuJoCo continuous-control benchmarks and remains the methodological reference for subsequent adversarial-RL work.

Adjacent formulations vary the disturbance type. Tessler, Efroni and Mannor (2019) introduce action-robust RL, where the adversary perturbs the protagonist’s executed action rather than system dynamics, with convergence guarantees in the tabular case and empirical transfer to continuous control [51]. An adversary perturbing the observation space (as required for an in-simulator spoofing analogue under replay) is structurally closer to RARL’s disturbance formulation than to action perturbation, since spoofing does not contest the protagonist’s action interface.

Bansal et al. (2018) establish that competitive multi-agent training under self-play in simple environments produces behaviour substantially more complex than the environment, providing a foundational result for self-play as an emergent-curriculum mechanism in zero-sum games [52]. The pattern carries over to settings far from their humanoid sumo benchmark: training against an adversary that learns alongside the protagonist produces increasingly capable adversaries by construction.

Gleave et al. (2020) identify the structural limit: a policy trained via self-play can remain brittle to novel adversarial strategies because self-play does not explore the full space of adversarial behaviour, and adversaries can be constructed that defeat state-of-the-art self-play victims with less than 3% of the victim’s training timesteps [53]. The Pinto–Bansal–Gleave sequence captures the basic dialectic: co-training produces measurable robustness gains, but bounded by adversary-strategy diversity at training time. Documented mitigations like league-based training, population methods, diverse-prior initialisations address this at substantial compute cost.

**Robust Markov decision processes.** A complementary paradigm addresses environment-model misspecification through robust Markov decision processes rather than co-trained adversaries. Nilim and El Ghaoui (2005) and Iyengar (2005) independently established the framework, in which the agent optimises worst-case expected return over an uncertainty set of transition kernels rather than a single nominal model, with the key tractability result that under a state-action rectangularity assumption the problem decomposes into a Bellman-style recursion of essentially non-robust complexity [54], [55]. Wiesemann, Kuhn and Rustem (2013) generalise to data-driven uncertainty sets estimated from observed transitions [56].

Robust MDPs and adversarial co-training address the same misspecification-robustness problem through different mechanisms. Robust MDPs avoid the adversary-overfitting failure mode of Gleave et al., since the agent is optimised against an uncertainty set rather than a specific opponent; the cost is conservatism: an over-large set produces pessimistic policies, an under-small set excludes deployment-time disturbances. Adversarial co-training avoids conservatism but inherits adversary overfitting. Robust MDPs also require explicit uncertainty-set characterisation, which in the market-making-under-spoofing setting would need to span parametric dynamics and structural adversary-behaviour uncertainty; a specification problem the literature has not addressed.

## 4.2 Adversarial training for market making

Spooner and Savani (2020) are the primary methodological precedent for adversarial market making [8]. They frame robust market making as a two-player zero-sum stochastic game in which a market maker and an adversary are co-trained, with the adversary controlling parameters of the underlying A–S dynamics (drift, order-arrival intensities, and order-arrival decay coefficients) so as to minimise the market maker’s cumulative reward. The formulation directly adapts Pinto et al.’s RARL to the financial setting, with A–S parameters replacing the disturbance forces of the original. Their results demonstrate measurable gains across standard performance metrics including Sharpe ratio over non-adversarial baselines.

Three properties warrant examination in light of the spoofing problem. First, the adversary is a parametric perturber addressing what the authors term “epistemic risk”; uncertainty about market-model parameter values at deployment, so it is positioned against parametric model misspecification rather than against an order-placing adversary that manipulates the visible LOB. An adversary that perturbs market parameters has no order-level cost structure to which the accidental-fill cost or regulatory-penalty intensity of a real spoofer, as characterised by Cartea et al. (2020), could meaningfully attach. Second, the market maker’s state vector contains only time and inventory; there is no detection channel through which the policy could discount a

suspicious LOB configuration. Third, the dynamics’ volatility coefficient is held fixed across training, so the agent cannot condition on regime shifts of the kind documented by Brogaard et al. (2018).

Components of the proposed configuration have appeared individually — Cartea, Chang and García-Arenas (2023) study emergent spoofing by a learning market maker but treat manipulator and victim as the same agent [46]; Wang and Wellman (2020) treat detection as an external classifier with no embedding in any market-making policy [48]; the signal-enriched DRL line of Section 2.2 operates without manipulation exposure. None combine an external order-placing adversary, an embedded detection channel consumed by the policy at inference, and explicit regime conditioning.

### 4.3 Auxiliary objectives and multi-task learning in RL

Embedding a spoofing-detection signal within a market maker’s policy network is, formally, an auxiliary-task RL problem: a primary policy-gradient objective is jointly optimised with a secondary loss whose gradient updates a shared representation. The canonical reference is the UNREAL agent of Jaderberg et al. (2017), augmenting a base A3C agent with auxiliary objectives including pixel control (auxiliary policies maximising pixel-intensity change across image regions), reward prediction, and value-function replay [57]. The empirical finding (auxiliary tasks substantially improve sample efficiency and final performance, with the largest gains under sparse reward) has been replicated. Shelhamer et al. (2017) frame the same construction from a self-supervision perspective, showing that joint optimisation with self-supervised auxiliary losses constructed from states, actions, and successors improves data efficiency without external annotation [58].

The auxiliary objectives in these works are themselves *self-supervised*: pseudo-rewards from the agent’s own experience, no external labelling. The proposed spoof-detection auxiliary task differs structurally: detection labels must come from an oracle that knows which order-book events were generated by the adversary versus the replay. The auxiliary task is supervised while the primary task remains policy-gradient RL. This combination, a supervised auxiliary loss with a non-stationary RL primary loss over a shared encoder, is not characterised in either the (overwhelmingly self-supervised) auxiliary-task RL literature or the (overwhelmingly supervised-on-both-sides) MTL literature below.

The supervised multi-task learning literature begins with Caruana (1997), who established that jointly training a shared representation on related tasks improves sample efficiency and generalisation through an implicit regularisation effect, typically via hard parameter sharing (shared encoder, task-specific heads) [59]. Yu et al. (2020) introduce Gradient Surgery (PCGrad), projecting conflicting gradients to eliminate interfering components, on the basis that gradient interference — two tasks producing gradients pointing in opposing directions over shared encoder parameters — is real and measurable [60]. Formally, for task gradients  $g_i, g_j$ , if  $\langle g_i, g_j \rangle < 0$ , PCGrad projects  $g_i$  onto the normal plane of  $g_j$ :

$$g_i^{\text{PC}} = g_i - \frac{\langle g_i, g_j \rangle}{\|g_j\|^2} g_j. \quad (4)$$

The shared encoder is updated using  $\sum_i g_i^{\text{PC}}$  rather than  $\sum_i g_i$ . The Caruana-to-Yu trajectory refines the original claim: where Caruana established that MTL can help, Yu et al. establish that whether it helps depends on gradient compatibility, and naive joint training can underperform single-task baselines when interference is severe.

For this pairing (quoting as primary task, spoofing detection as auxiliary) the two are plausibly compatible: LOB features distinguishing spoofed from bona-fide flow are features a robust

market maker should attend to, and detection accuracy and quoting quality are not obviously in tension. Compatibility cannot be assumed, however: the auxiliary task is trained from a static label distribution (the replay’s adversary indicator), while the primary task’s effective state distribution shifts as the policy improves; the interaction over a shared encoder is the characterisation gap noted above. Empirical defaults for this configuration are unclear: gradient-balancing methods (PCGrad, gradient-magnitude normalisation) are candidates, but none has been evaluated for an adversarial RL primary plus a supervised auxiliary detection head.

## 5 Synthesis: collective state of the field

The four literatures have developed largely in parallel, and what they collectively establish, contest, and leave open can be summarised in turn.<sup>1</sup>

**What is established.** The microstructure literature has converged on inventory risk and adverse selection as the two fundamental risks for market makers, with Glosten–Milgrom and Kyle as the theoretical channels and PIN/VPIN as contested empirical instantiations. Order-flow imbalance, including cancellations, is an approximately linear depth-scaled predictor of short-horizon price change. Rapid cancellation is prevalent. Volatility is regime-switching, and liquidity provision is regime-dependent. The DRL market-making literature has established that learned agents match or exceed A–S baselines benignly, with PPO the algorithmic default and both replay-based (JAX-LOB, JaxMARL-HFT) and agent-based (ABIDES, PyMarketSim) simulators in active use. The spoofing literature has characterised manipulation’s economic structure through optimal control, with separate work showing it can emerge from learning. Adversarial RL has established that co-training produces measurably more robust policies, with adversary overfitting and the robustness–performance trade-off as failure modes; robust MDPs provide a tractable alternative optimising worst-case return over an uncertainty set. Auxiliary-task RL has established that self-supervised auxiliaries improve sample efficiency through shared-representation effects; supervised MTL has documented gradient interference and negative transfer as real risks.

**What is contested or unresolved.** The applicability of the CKS impact coefficient to specifically manipulative cancellation flow — versus the aggregate cancellation population on which it was estimated — is unknown. Whether a representation trained to detect spoofing can outperform a PIN/VPIN-style aggregate signal in the spoofing subpopulation has not been tested. The A–S exponential order-arrival assumption has not been systematically tested against tick-constrained markets. The PPO-versus-off-policy choice under adversarial non-stationarity is a convention, not a settled finding. Whether alternating-freeze training produces stable policies in market-maker-versus-attacker games is unverified. The relative merits of robust-MDP and adversarial-co-training approaches in the financial setting have not been directly compared, and uncertainty-set specification for this setting is open. The false-positive behaviour of detectors trained on prosecuted cases when applied to full order streams is uncharacterised. The interaction between a non-stationary RL primary loss and a supervised auxiliary loss over a shared encoder is uncharacterised in either literature. The regime-indicator choice (realised volatility, VIX, filtered

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<sup>1</sup>The literature was assembled via Google Scholar, Semantic Scholar, and arXiv, using keyword combinations from the four constituent literatures: `market making with reinforcement learning`; `adversarial reinforcement learning` and `robust reinforcement learning`; `robust Markov decision process`; `limit order book with spoofing, layering, or manipulation`; `order flow imbalance`; `probability of informed trading`; `regime switching with volatility`; `auxiliary task with reinforcement learning or multi-task learning`; and `market simulator with agent-based or limit order book`. Date range: 1981–2025. Inclusion prioritised peer-reviewed venues and preprints with substantial downstream citation; enforcement-action references draw on SEC and federal court documents directly. The negative existential claim in the final paragraph is to be read as conditional on this search scope.

Markov-switching probabilities, BOCPD run-length posteriors) for conditioning a high-frequency policy is unsettled.

**What is structurally constrained by simulator choice.** The most consequential structural constraint cuts across the DRL and spoofing literatures: market-replay simulators (JaxMARL-HFT, JAX-LOB) cannot produce endogenous price response to injected orders, so adversarial training in such simulators tests robustness to state perturbation rather than causal spoofing, and any economic cost model attached to the adversary must be reconciled with this constraint. The agent-based alternative (ABIDES in IABS mode, PyMarketSim) requires a calibrated agent population on which no consensus exists and incurs substantial throughput costs. This constraint is acknowledged in the simulator papers but has not been systematically addressed in subsequent adversarial-training work.

**What no prior work has jointly examined.** Within these established findings and unresolved questions, no prior work has jointly examined: an economically-constrained adversarial agent whose perturbations are structurally derived from the operational profile of spoofing, with per-action costs grounded in the Cartea–Jaimungal–Wang cost-model lineage; spoof detection embedded as a supervised auxiliary objective within a DRL market maker’s policy network, with the detection head’s output consumed by the policy at inference; and explicit conditioning on a market volatility regime indicator, motivated by the regime-dependent depth and liquidity-provision dynamics documented in the microstructure literature. The individual components (adversarial co-training, supervised auxiliary objectives, contextual RL with regime indicators) are established techniques. The combination is motivated structurally rather than by convenience: the adversary generates the training distribution under attack; the detection head provides a representational channel through which the policy can condition on attack-specific features at inference; the regime indicator modulates that response across market states whose depth dynamics (and hence the relationship between depth signals and price impact) are themselves non-stationary (Brogaard et al., 2018). Each component addresses a failure mode the others leave open: adversarial training without a detection channel produces an undifferentiated response to manipulation; detection without an active adversary trains the channel against an unrepresentative attack distribution; and regime conditioning without either lacks the manipulation-aware signal it would modulate. This project tests the hypothesis that the jointly integrated system of those components produces a statistically significant improvement in risk-adjusted PnL under observation-space perturbation relative to a non-adversarially-trained PPO baseline, without statistically significant degradation in inventory-adjusted PnL under unperturbed replay conditions. Secondary hypotheses are that the detection head achieves above-chance classification accuracy on held-out adversarial episodes, and that regime-conditioning reduces performance variance across high- and low-volatility periods.

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